# White Blood Cells

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## Identity Sheet

| Property | Value |
|---|---|
| **Name** | White Blood Cells (Leukocytes) |
| **Other names** | WBCs, leukocytes, white cells |
| **Nature** | Nucleated immune cells |
| **Size** | 10–15 µm (neutrophils), 7–10 µm (lymphocytes), up to 20 µm (monocytes) |
| **Predominant type in urine** | Neutrophils (~95% of urinary WBCs) |
| **Normal blood count** | 4,000–11,000/µL |

**Morphology:**

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Neutrophil: (2-5 lobed nucleus, granular cytoplasm,  12 um)
Lymphocyte: (round nucleus, scant cytoplasm,  8 um)
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White blood cells in urine (pyuria) indicate an inflammatory process in the urinary tract. Neutrophils are the dominant type, recruited to the site of infection or inflammation. WBCs are distinguished from RBCs by their larger size, visible nucleus, and granular cytoplasm. WBC casts (tubular casts containing WBCs) indicate renal parenchymal inflammation (pyelonephritis, interstitial nephritis).

### Molecules Not to Be Confused With

| Entity | Description | Key Difference |
|---|---|---|
| **Red blood cells** | Anucleate, 6–8 µm, biconcave | Smaller, no nucleus, no granules |
| **Renal tubular epithelial cells** | 12–20 µm, round nucleus | Slightly larger; single large round nucleus; granular cytoplasm |
| **Transitional epithelial cells** | 20–40 µm, variable shapes | Much larger, variable morphology |
| **Bacteria** | 0.5–3 µm | Much smaller; no visible internal structure by light microscopy |

## Medical Information

### Origin

***Endogenous: Haematopoiesis*** WBCs are produced in the **bone marrow** from haematopoietic stem cells. Neutrophils (the most relevant urinary WBC) mature over ~14 days and circulate for 6–10 hours before migrating to tissues. In urinary tract inflammation, neutrophils are recruited by chemokines (IL-8, complement C5a) from blood vessels in the bladder wall, ureter, or renal parenchyma, passing through the urothelium into the urine.

***Exogenous*** No exogenous source. WBCs in urine originate exclusively from the patient's immune system responding to local inflammation or infection.

## Primary & Secondary Biological Roles

**Primary role:** - **Immune defence:** Neutrophils are first responders to bacterial infection; they phagocytose and kill pathogens through oxidative burst (reactive oxygen species), degranulation (enzymes: myeloperoxidase, elastase), and neutrophil extracellular traps (NETs).

**Secondary roles:** - **Inflammation mediators:** Release cytokines (TNF-α, IL-1β) and lipid mediators (leukotrienes, prostaglandins) that amplify inflammatory responses. - **Tissue damage markers:** Eosinophils in urine (rare) suggest allergic interstitial nephritis or parasitic infection.

## Catabolism and Elimination Pathway

- Neutrophils in urine undergo **apoptosis** and lysis within hours.
- Enzymes released during lysis (leukocyte esterase) can be detected by dipstick even after cells disintegrate.
- In the body, apoptotic neutrophils are cleared by macrophages (efferocytosis).
- Urinary WBCs are eliminated by voiding.

## Expression in Humans

### *Normal Levels*

| Compartment | Reference Range |
|---|---|
| **Blood WBC count** | 4,000–11,000/μL |
| **Urine WBCs (microscopy)** | 0–5 WBC/HPF (high-power field) |
| **Urine WBCs (flow cytometry)** | <25 WBC/μL (Sysmex UF) |
| **Dipstick leukocyte esterase** | Negative |

***Factors Influencing Levels*** **Increased WBCs in urine (pyuria):** - Urinary tract infection (most common cause) - Pyelonephritis - Interstitial nephritis (drug-induced, allergic) - Glomerulonephritis - Renal transplant rejection - Urolithiasis - Bladder cancer - Vaginal/cervical contamination - Prostatitis

**Decreased / absent (normal):** - Healthy urinary tract - No inflammation or infection - Immunosuppression (may reduce pyuria despite infection)

### *Associated Pathologies*

| Condition | WBC Pattern | Key Symptoms |
|---|---|---|
| **Uncomplicated cystitis** | >10 WBC/HPF + bacteria | Dysuria, frequency, urgency |
| **Pyelonephritis** | WBCs + WBC casts + bacteria | Fever, flank pain, nausea |
| **Sterile pyuria** | WBCs without bacteria on culture | TB, interstitial nephritis, STIs, nephrolithiasis, renal cancer |
| **Drug-induced interstitial nephritis** | WBCs + eosinophils + WBC casts | AKI after drug exposure; eosinophiluria pathognomonic |
| **Renal transplant rejection** | WBCs + tubular epithelial cells | Rising creatinine post-transplant |

## Presence in Urine

**Should they be normally present?** In **trace amounts** only — up to 5 WBC/HPF is considered normal. Greater numbers define pyuria.

**Normal urinary levels:** <5 WBC/HPF; <25 WBC/μL.

**Form in urine:** **Intact nucleated cells**, predominantly neutrophils. May degenerate into "glitter cells" (swollen neutrophils in hypotonic urine with brownian motion of granules). WBC casts indicate renal origin.

**Pathological significance:**

| Urinary WBCs | Possible Causes | Prevalence |
|---|---|---|
| **>5 WBC/HPF (pyuria)** | UTI, pyelonephritis, interstitial nephritis, STI | UTI alone: ~150 million/year globally |
| **Sterile pyuria** | TB (~5–10% of sterile pyuria), drugs, stones, cancer | Varies |
| **Normal** | Healthy | — |

**Solubility:** Not applicable (cells).

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## Detection in Urine

### Available Clinical Assays

1. **Dipstick leukocyte esterase (LE):**
   - **Principle:** Esterase enzyme released from lysed neutrophils hydrolyses an indoxyl ester substrate → indoxyl, which reacts with a diazonium salt → violet azo dye.
   - **Detection:** Colour change graded negative, trace, 1+, 2+, 3+.
   - **LOD:** ~10–25 WBC/μL (~5–15 WBC/HPF).
   - **Absorption:** ~550 nm (violet dye).
   - **Sensitivity for UTI:** 75–90%. **Specificity:** 65–80%.
   - **Advantages:** Rapid, cheap, detects lysed cells.
   - **Disadvantages:** False positives from contamination; cannot count cells.
2. **Urine sediment microscopy:**
   - **Principle:** Centrifuged urine sediment examined at 400x. WBCs identified by size (~12 μm), visible nucleus, granules.
   - **LOD:** ~1 WBC/HPF.
   - **Advantages:** Counts cells, identifies casts, distinguishes cell types.
   - **Disadvantages:** Operator-dependent, time-consuming.
3. **Automated flow cytometry (Sysmex UF-5000):**
   - **Principle:** Fluorescent nucleic acid dye labels WBC DNA → fluorescence + scatter. WBCs have higher fluorescence than RBCs (which lack nuclei).
   - **Detection:** Ex 488 nm; WBCs show strong fluorescence (nucleated).
   - **LOD:** ~5 WBC/μL.
   - **Advantages:** Rapid, objective, reproducible.
   - **Disadvantages:** Cannot identify WBC subtypes or casts.

### Optimal Urine Type for Measurement

- **Midstream clean-catch urine** — standard specimen.
- Analyse within **1–2 hours** (WBCs lyse rapidly in hypotonic/alkaline urine).
- **First morning urine** provides higher concentration.
- Avoid vaginal contamination in women (instruct proper collection technique).
- LE test remains positive even after WBC lysis (detects released enzyme).

## Actual Gold Standard

**Manual microscopy** of centrifuged urine sediment at 400x remains the reference for WBC quantification and morphological assessment (including cast identification). Automated analysers (Sysmex UF) are increasingly used for screening with reflex to microscopy for abnormal results.

## Interferences in Measurement

| Interference | Effect | Mechanism |
|---|---|---|
| **Vaginal contamination** | False positive | Vaginal leukocytes, epithelial cells |
| **Oxidising agents (hypochlorite)** | False positive (LE dipstick) | Chemical oxidation of substrate |
| **Formaldehyde** | False positive (LE) | Non-specific colour reaction |
| **High glucose (>3 g/dL)** | False negative (LE) | Inhibits enzymatic reaction |
| **High albumin (>500 mg/dL)** | False negative (LE) | Enzyme inhibition |
| **Ascorbic acid** | Possible false negative | Antioxidant interference (minor) |
| **Tetracycline, cephalexin** | False negative (LE) | Drug inhibition of esterase activity |
| **Delayed analysis** | Decreased WBC count (microscopy) | Cell lysis in old specimen |

## Research Detection Methods

### *Spectroscopy Detection (UV-Vis / NIR)*

- **UV-Vis:** No direct spectroscopic method for WBCs. Myeloperoxidase (MPO) from neutrophils has a Soret-like absorption at **430 nm**, but concentrations in urine are too low for direct detection.
- **NIR:** Not applicable for WBC detection.
- **Turbidimetry:** Urine turbidity correlates loosely with high WBC and bacterial counts, but is non-specific.

### *Fluorescence Detection*

- **Nucleic acid staining (flow cytometry):** Polymethine/acridine orange dyes bind WBC DNA/RNA. Ex 488 nm / Em 520 nm (DNA) and Em >600 nm (RNA). LOD: ~5 WBC/µL.
- **Immunofluorescence:** CD45-FITC antibody staining for pan-leukocyte marker. Ex 490 / Em 520 nm. LOD: ~1 WBC/µL. Specific for leukocytes.
- **Calcein-AM viability staining:** Live WBCs convert calcein-AM to fluorescent calcein. Ex 495 / Em 515 nm. LOD: ~10 WBC/µL.
- **Sample prep:** For flow cytometry, no prep; for immunofluorescence, fixation + antibody incubation.

***Raman Detection*** WBC Raman spectra reflect their biomolecular composition:

| Peak ($cm^{-1}$) | Assignment |
|---|---|
| **~720–730** | Adenine (DNA) |
| **~780** | Cytosine/uracil (nucleic acids) |
| **~1003** | Phenylalanine |
| **~1340** | $CH_2$ wagging (proteins) |
| **~1450** | $CH_2/CH_3$ deformation |
| **~1660** | Amide I |

- Single-cell Raman profiling is possible with confocal Raman (Ex 532 or 785 nm). Can distinguish neutrophils, lymphocytes, and monocytes by spectral differences.

- LOD: Single cell.
- **Sample prep:** Concentration by centrifugation; cell transfer to optical substrate.

#### *FTIR Detection*

| Band ($cm^{-1}$) | Assignment | Notes |
|---|---|---|
| **~1080** | $PO_2^-$ symmetric stretch (DNA) | Higher DNA content than RBCs |
| **~1240** | $PO_2^-$ asymmetric stretch (DNA) | — |
| **~1540** | Amide II | Proteins |
| **~1650** | Amide I | Proteins |
| **~2850–2960** | C-H stretches (lipids) | Membrane lipids |

- ATR-FTIR on dried sediment can detect WBCs at high concentrations (>10^4/μL). Not routinely used.
- Can potentially distinguish infected vs non-infected urine by overall spectral profile.
- **Sample prep:** Centrifuge, wash, dry on ATR crystal.

#### *Voltammetry Detection*

- **Myeloperoxidase (MPO) electrochemistry:** MPO from neutrophils can be detected by its peroxidase activity on modified electrodes. LOD: ~0.1 ng/mL MPO.
- **Impedance cytometry:** Microfluidic impedance sensing for single-cell counting and sizing. LOD: ~10 cells/μL.
- **Sample prep:** Minimal for impedance; cell lysis for MPO.

### Other Detection Technologies

1. **Automated digital microscopy (Iris iQ200, Sysmex UN):** AI-based image classification of urine sediment. LOD: ~5 WBC/μL. Automated, objective.
2. **Smartphone microscopy:** Portable imaging of urine sediment. LOD: ~10–50 WBC/μL. Low cost.
3. **Lateral flow for MPO/LE:** Paper-based strips for leukocyte marker detection. LOD: comparable to dipstick.
4. **PCR for 16S rRNA (indirect):** Detects bacterial cause of pyuria. Not WBC-specific.

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## Summary Table: Detection Methods Comparison

| Method | LOD | Wavelength / Key Parameter | Sample Prep | Strengths | Limitations |
|---|---|---|---|---|---|
| **Dipstick LE** | ~10–25 WBC/μL | ~550 nm (violet dye) | None | Rapid, cheap, detects lysed cells | Non-specific, false positives |
| **Microscopy** | ~1 WBC/HPF | 400x light microscopy | Centrifugation | Gold standard, morphology | Operator-dependent |
| **Flow cytometry (Sysmex)** | ~5 WBC/μL | Ex 488 nm / scatter | None | Automated, fast | No subtyping |
| **Immunofluorescence (CD45)** | ~1 WBC/μL | Ex 490 / Em 520 nm | Fixation, antibody | Specific for WBCs | Labour-intensive |
| **Single-cell Raman** | Single cell | Ex 532/785 nm; 1003 $cm^{-1}$ | Concentrate | Cell-type identification | Specialised |

| Method | LOD | Wavelength / Key Parameter | Sample Prep | Strengths | Limitations |
|---|---|---|---|---|---|
| **FTIR** | >10^4/μL | 1080, 1650 $cm^{-1}$ | Dry pellet | Multianalyte | Low sensitivity |
| **Impedance cytometry** | ~10/μL | Impedance | Microfluidic | Label-free, real-time | Prototype stage |
| **Digital microscopy (AI)** | ~5 WBC/μL | Digital imaging | None/auto | Automated, objective | Instrument cost |

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## References


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